

INFORMATION TECHNOLOGY

GRADE 12 · PAPER 2 THEORY

The Claude NSC Study Companion

Terminology · Exam Traps · Database Design · UML · Abbreviations

Aligned to the CAPS (Curriculum and Assessment Policy Statement) curriculum

Terms marked ★ appear most frequently in past NSC (National Senior Certificate) papers

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Part 1 — Rapid Fire Glossary

How to use this section

Terms marked with ★ appear in NSC Paper 2 year after year. If time is short, drill those first.

Where a term is an abbreviation, the full form is given in brackets. Learn both. Questions regularly ask what an abbreviation stands for, and a marker cannot award a mark for "IP address" when the question asked what IP stands for.

Cover the right-hand column with a ruler, say the definition aloud, then check. Anything you get wrong twice goes onto a separate page — that page becomes your final week of revision.

Systems Technologies

Term	Definition
★ CPU (Central Processing Unit)	The chip that fetches, decodes and executes instructions.
★ Core	An independent processing unit inside the CPU; more cores means more genuinely parallel work.
★ Cache (L1, L2, L3 — Level 1, 2 and 3)	Tiny, ultra-fast memory on the CPU holding data it is likely to need next. L1 is the smallest and fastest.
Clock speed	The number of instruction cycles per second, measured in GHz (Gigahertz).
★ RAM (Random Access Memory)	Volatile working memory; its contents are lost when power is switched off.
ROM (Read-Only Memory) / firmware	Non-volatile memory holding permanent start-up instructions.
★ Virtual memory	Disk space used as an extension of RAM when physical RAM runs out.
★ Thrashing	The system slows to a crawl because it spends more time swapping pages between RAM and virtual memory than doing useful work.
Buffer	A temporary memory area that smooths out speed differences between devices.
★ Spooling	Queueing jobs (usually printing) to disk so the CPU is not held up waiting.
Interrupt	A signal telling the CPU to pause its current task and attend to something urgent.
Bus	The pathway carrying data between components; its width affects throughput.
★ SSD (Solid State Drive) vs HDD (Hard Disk Drive)	Flash memory with no moving parts, fast and durable vs spinning magnetic platters, cheaper per GB (Gigabyte).
GPU (Graphics Processing Unit)	A specialised processor for graphics and highly parallel calculations.
★ Driver	Software that allows the operating system to communicate with a specific hardware device.
BIOS (Basic Input/Output System) / UEFI (Unified Extensible Firmware Interface)	Firmware that starts the machine, tests the hardware and hands control to the operating system. UEFI is the modern replacement for BIOS.
★ OS (Operating System)	System software that manages hardware, memory, processes, files and the user interface.
Multitasking	The operating system switching rapidly between processes so that they appear to run simultaneously.

Term	Definition
★ Utility software	Maintenance tools such as disk defragmenters, disk cleanup, antivirus and backup software.
★ Virtualisation	Running several virtual machines on one physical host by means of a hypervisor.
Embedded system	A dedicated computer built into a larger device, e.g. a washing machine or a car engine controller.
★ IoT (Internet of Things)	Everyday objects fitted with sensors and network connectivity that collect and exchange data.
RFID (Radio Frequency Identification) / NFC (Near Field Communication)	Short-range wireless tag reading; NFC is the very short-range "tap" version used for payments.
Biometrics	Authentication using unique physical traits such as fingerprints, iris patterns or facial features.
★ UPS (Uninterruptible Power Supply)	A battery backup giving you time to save work and shut down cleanly during a power failure.
★ Ergonomics	Designing the workspace and equipment to fit the user and prevent strain injuries.
★ Green computing	Reducing the environmental impact of ICT (Information and Communication Technology) across the whole life cycle of a device.
★ e-waste (electronic waste)	Discarded electronic equipment containing toxic materials that requires specialised disposal.
Convergence	Several separate functions merging into one device, e.g. a phone that is also a camera and a GPS (Global Positioning System) receiver.
Bloatware	Unwanted pre-installed software that consumes storage and processing resources.
Plug and play	The operating system automatically detecting and configuring newly connected hardware.
Throughput	The actual amount of useful work completed per unit of time, as opposed to theoretical speed.

Software Concepts

Term	Definition
System vs application software	Software that runs and manages the computer vs software that performs a task for the user.
★ Open source	Software whose source code is freely available to view, modify and redistribute.
★ Proprietary software	Closed-source software that is owned and licensed under restrictive terms.
Freeware / shareware	Free to use permanently vs free to try for a period, after which payment is required.
★ EULA (End User Licence Agreement)	The contract governing how the software may legally be used.
Patch / update / upgrade	A small fix, a broader improvement release, and a move to a major new version.
★ Compiler vs interpreter	Translates the entire program once into machine code vs translates and executes one line at a time.

Term	Definition
★ DRM (Digital Rights Management)	Technology and access controls used by copyright holders to restrict how digital content may be copied, shared, printed or used after purchase.
Product / licence key	A code that activates a legal copy of software and is a common form of DRM.
Digital watermark	Hidden identifying data embedded in a file so that illegal copies can be traced back to their source.
IDE (Integrated Development Environment)	Software combining an editor, compiler, debugger and form designer in one package, e.g. Delphi.

Communication and Network Technologies

Term	Definition
★ LAN (Local Area Network)	A network covering a single site such as a school or an office building.
★ WAN (Wide Area Network)	A network spanning cities, countries or continents. The internet is the largest WAN.
PAN (Personal Area Network)	A network covering a few metres around one person, typically using Bluetooth.
MAN (Metropolitan Area Network)	A network covering a city or a large campus.
Node	Any addressable device connected to a network.
★ Client-server vs peer-to-peer (P2P)	A central server supplies resources to clients vs every machine both provides and consumes resources.
Topology	The physical or logical layout of a network: star, bus, mesh or ring.
NIC (Network Interface Card)	The hardware that gives a device its network connection and its MAC address.
Switch	A device that forwards frames only to the intended port on a LAN.
★ Router	A device that forwards packets between different networks and finds the best path.
Modem (modulator-demodulator)	Converts between digital data and the signal that the physical line can carry.
WAP (Wireless Access Point)	A device that provides Wi-Fi connectivity into a wired network.
★ Bandwidth	The maximum amount of data that can be transferred per second, measured in Mbps (Megabits per second).
★ Data cap	A contractual limit on the total volume of data that may be used in a period.
★ Shaping / throttling	The ISP (Internet Service Provider) prioritising certain traffic types, or deliberately reducing speed.
Latency	The delay before data begins to arrive at its destination.
Attenuation	The weakening of a signal as it travels over distance.
★ UTP (Unshielded Twisted Pair) / fibre-optic / coaxial	Copper twisted pair / glass carrying light, fastest and immune to interference / shielded copper.
★ Wi-Fi	The standard for wireless local area network connections. Note that Wi-Fi is a trade name and is not an abbreviation of anything.
Bluetooth	A short-range wireless standard for pairing devices within a personal area network.
★ LTE (Long Term Evolution) / 5G (fifth generation)	Successive generations of high-speed cellular broadband.

Term	Definition
★ Protocol	An agreed set of rules governing how devices communicate.
★ TCP/IP (Transmission Control Protocol / Internet Protocol)	The core protocol suite of the internet: TCP guarantees reliable delivery, IP handles addressing and routing.
★ HTTP (Hypertext Transfer Protocol)	The protocol used to transfer web pages between a server and a browser.
★ HTTPS (Hypertext Transfer Protocol Secure)	HTTP with an added layer of TLS (Transport Layer Security) encryption.
FTP (File Transfer Protocol)	The protocol used to upload files to and download files from a server.
★ SMTP (Simple Mail Transfer Protocol)	The protocol used to send outgoing e-mail.
★ POP3 (Post Office Protocol version 3)	Downloads e-mail to one device and normally removes it from the server.
★ IMAP (Internet Message Access Protocol)	Keeps e-mail on the server and synchronises it across all devices.
★ DNS (Domain Name System)	The service that translates human-readable domain names into IP addresses.
DHCP (Dynamic Host Configuration Protocol)	The service that automatically issues IP addresses to devices joining a network.
★ IP (Internet Protocol) address	A logical, changeable address identifying a device on a network.
★ MAC (Media Access Control) address	A permanent hardware address burnt into the NIC and unique worldwide.
Packet switching	Data is split into packets that are routed independently and reassembled at the destination.
★ VPN (Virtual Private Network)	An encrypted tunnel over a public network that creates a private, secure connection.
★ Cloud computing	On-demand computing resources hosted remotely and accessed over the internet.
★ SaaS (Software as a Service)	Renting ready-made software that runs in the cloud, e.g. an online office suite.
★ PaaS (Platform as a Service)	Renting a hosted platform on which to develop and run your own applications.
★ IaaS (Infrastructure as a Service)	Renting raw servers, storage and networking in the cloud.
Grid computing	Many networked computers pooled together to work on one large problem.
Tethering / hotspot	Sharing one device's mobile data connection with other devices.
BYOD (Bring Your Own Device)	A policy allowing employees to use their own personal devices on the organisation's network.

Security and Threats

Term	Definition
★ Firewall	Hardware or software that filters traffic between networks according to security rules.
★ Encryption	Scrambling data so that only someone holding the correct key can read it.
★ Digital certificate	Proof issued by a trusted third party that a website or sender is genuine.

Term	Definition
Digital signature	An encrypted stamp verifying the sender's identity and that the content has not been altered.
★ 2FA (Two-Factor Authentication) / MFA (Multi-Factor Authentication)	Requiring two or more independent proofs of identity, e.g. a password plus a one-time PIN (Personal Identification Number).
★ Malware (malicious software)	An umbrella term for any software written with harmful intent.
Virus	Malicious code that attaches itself to a host file and requires a user action to spread.
★ Worm	Self-replicating malware that spreads across networks without any user intervention.
★ Trojan (Trojan horse)	Malware disguised as legitimate, useful software.
★ Ransomware	Malware that encrypts the victim's files and demands payment for the decryption key.
★ Spyware / keylogger	Software that secretly monitors activity / that records every keystroke typed.
Rootkit	Malware that embeds itself deep in the operating system to avoid detection.
★ Phishing	Fraudulent messages that trick the user into surrendering personal details or credentials.
★ Pharming	Redirecting a user to a fake website even when the correct address has been typed.
Spoofing	Faking a source address or identity to appear as someone or something else.
★ Social engineering	Manipulating people rather than systems in order to gain unauthorised access.
★ Bot / botnet / zombie	An infected machine / a network of infected machines / one individual compromised host.
★ DDoS (Distributed Denial of Service) attack	Overwhelming a server with traffic from many sources until it can no longer respond.
Cookie	A small text file a website stores on your device to remember information about you.
Hoax	A false warning or story circulated by well-meaning users.
Adware	Software that displays unwanted advertising, often bundled with free downloads.
Zero-day exploit	An attack that uses a vulnerability before the vendor has released a fix.

Internet and e-Communication

Term	Definition
★ Web 2.0	The user-generated, interactive and collaborative web.
Blog (weblog) / vlog (video blog)	Chronological written posts / the same idea in video form.
Podcast / wiki	A series of downloadable audio episodes / a collaboratively edited set of web pages.
★ RSS (Really Simple Syndication)	A feed format that automatically pushes new content to subscribers.

Term	Definition
★ VoIP (Voice over Internet Protocol)	Voice calls carried over the internet instead of the traditional telephone network.
★ Search engine	A site that indexes web pages and returns ranked results for a user query.
★ Crawler / spider / bot	A program that follows links to gather web pages for a search engine index.
URL (Uniform Resource Locator)	The full address of a specific resource on the web.
e-commerce / EFT (Electronic Funds Transfer) / e-wallet	Trading online / an electronic bank transfer / stored digital value on a device.
Netiquette (network etiquette)	The accepted standards of polite and responsible behaviour online.
★ Digital divide	The gap between those with meaningful access to ICT and those without it.
Information overload	Being overwhelmed by more information than can reasonably be processed.
★ Responsive design	Web layout that adapts fluidly to any screen size or device.
★ Streaming	Playing media as it arrives over the network rather than downloading it first.
CMS (Content Management System)	Software that lets non-programmers publish and maintain website content.
Bookmark / favourite	A saved link allowing quick return to a page.
FAQ (Frequently Asked Questions)	A page answering the questions users most commonly ask.

Data and Information Management

Term	Definition
★ Data vs information	Raw, unprocessed facts vs data that has been processed and given meaning and context.
★ GIGO (Garbage In, Garbage Out)	Flawed input data guarantees flawed output, no matter how good the processing is.
★ Database	An organised collection of related data.
★ DBMS (Database Management System)	The software layer between the user and the database that controls all access to the data.
Table / record / field	The data for one entity / one row describing one instance / one attribute column.
★ PK (Primary Key)	A field that uniquely identifies each record; it may never be null or duplicated.
★ FK (Foreign Key)	A field in one table that refers to the primary key of another table.
Composite key	A primary key made up of two or more fields used together.
★ Relationships (1:1, 1:M, M:M)	One-to-one / one-to-many / many-to-many, which must be resolved with a junction table.
★ Referential integrity	The rule that a foreign key may not refer to a record that does not exist.
★ Data integrity	Data being accurate, complete, consistent and current.
★ Data redundancy	The same data being unnecessarily stored in more than one place.
Data independence	Being able to change how data is stored without having to rewrite the applications that use it.

Term	Definition
★ Normalisation	Restructuring tables to remove redundancy and eliminate insert, update and delete anomalies.
★ Insert / update / delete anomaly	Problems that arise from poorly designed, unnormalised tables.
Entity	A thing about which data is stored, which becomes a table in the database.
★ ERD (Entity-Relationship Diagram)	A diagram showing the entities in a database and the relationships between them.
★ SQL (Structured Query Language)	The standard language for querying and manipulating relational databases.
Index	A structure that speeds up searching and sorting on a particular field.
★ Metadata	Data that describes data: field names, data types, sizes and creation dates.
★ Data warehouse	A large central store of historical data consolidated from many sources for analysis.
★ Data mining	The process of discovering hidden patterns and relationships in large data sets.
★ Big data	Data sets too large, too fast-moving or too varied for traditional tools (volume, velocity, variety).
OLAP (Online Analytical Processing)	Examining data-warehouse data from several dimensions at once.
★ Validation	Automated checks confirming that entered data is reasonable and in the correct format.
★ Verification	Checking that the data entered matches the original source exactly.
Backup vs archive	A copy kept for recovery vs long-term storage of data no longer in active use.
RAID (Redundant Array of Independent Disks)	Several drives combined to provide redundancy, improved speed, or both.
★ POPIA (Protection of Personal Information Act)	South African legislation governing the lawful collection, use and protection of personal information.
CSV (Comma-Separated Values)	A plain-text format in which each line is a record and commas separate the fields.
XML (eXtensible Markup Language)	A tagged, self-describing text format for storing and transporting structured data.
JSON (JavaScript Object Notation)	A lightweight text format for structured data, widely used by web services.
Atomic data	Data broken down into the smallest useful units, with one value per field.

Solution Development

Term	Definition
★ Algorithm	A finite, ordered set of unambiguous steps that solves a problem.
Pseudocode / flowchart	A structured-English plan / a diagrammatic plan of an algorithm.
★ Trace table	A table that tracks variable values line by line in order to test logic by hand.
★ Event-driven programming	A style in which code executes in response to user or system events.

Term	Definition
★ Object / object class	An instance with attributes and methods / the blueprint that defines them.
Instantiate	To create an object in memory from a class, which is done by calling the constructor.
★ Constructor	A method that creates an object and initialises its attributes.
Destructor	A method that frees an object from memory.
★ Encapsulation	Bundling data with the methods that operate on it and hiding the internals behind an interface.
★ Accessor (getter) method	A method that returns the value of a private attribute. It normally takes no parameter.
★ Mutator (setter) method	A method that changes the value of a private attribute. It must receive a parameter.
★ Auxiliary (helper) method	Any method that is not a constructor, an accessor or a mutator, e.g. toString or a calculation method.
★ Function vs procedure	Returns a single value vs performs an action and returns nothing.
★ Value vs reference parameter	A copy is passed and the original is safe vs the actual variable is passed and may be changed.
★ Local vs global variable	Visible only inside its own routine vs visible throughout the unit.
Scope	The region of code in which an identifier can be accessed. Private members have class scope only.
★ Array (one-dimensional / two-dimensional)	A fixed-size ordered collection of items of the same type stored under one name.
Parallel arrays	Separate arrays in which matching index positions describe the same entity.
Text file / delimiter	Plain-text data storage / the character used to separate fields in a record.
Linear search / bubble sort	Checking items one by one from the start / repeatedly comparing and swapping neighbours.
★ Syntax / logic / runtime error	Breaks the rules of the language / compiles but produces wrong output / crashes during execution.
★ Debugging	Systematically locating and removing errors from a program.
Exception handling	Catching runtime errors gracefully instead of allowing the program to crash.
★ Modular programming	Dividing a solution into independent, reusable routines.
★ SDLC (Software Development Life Cycle)	Analysis, design, development, testing, implementation and maintenance.
Prototype	An early working model built to gather user feedback before full development.
★ Alpha vs beta testing	Testing done in-house by the developers vs testing by real users outside the company.
★ User vs technical documentation	How to use the system, for the end user vs how it works, for the developer.
★ UML (Unified Modeling Language)	A standard notation for modelling software, used in Grade 12 to draw a class diagram.
★ HTML (HyperText Markup Language)	The markup language that structures the content of web pages.
CSS (Cascading Style Sheets)	The language that controls the presentation and layout of HTML content.

Term	Definition
GUI (Graphical User Interface)	An interface using windows, icons, menus and a pointer.
Robustness	A program's ability to cope with invalid input and unexpected conditions without crashing.

Social and Ethical Implications

Term	Definition
★ Software piracy	Copying, distributing or using software in breach of its licence.
Copyright	The creator's legal right to control the reproduction and distribution of their work.
★ DRM (Digital Rights Management)	The technological measures that enforce copyright by controlling what a purchaser may do with digital content.
★ Plagiarism	Presenting someone else's work or ideas as your own.
★ Ethics vs legality	What is morally right vs what the law actually permits or prohibits.
★ Identity theft	Stealing someone's personal details in order to impersonate them for gain.
Cyberbullying	Using digital communication to harass, threaten or humiliate someone.
★ ECT Act (Electronic Communications and Transactions Act)	South African legislation regulating electronic transactions, signatures and communications.
★ RSI (Repetitive Strain Injury)	Injury to muscles and tendons caused by prolonged repeated movement such as typing.
★ CVS (Computer Vision Syndrome)	Eye strain, headaches and blurred vision caused by extended screen use.
Assistive technology	Hardware or software that enables users with disabilities to use a computer.
Ubiquitous computing	Computing embedded so widely in the environment that it becomes invisible.
★ AI (Artificial Intelligence)	Machines simulating intelligent behaviour such as reasoning, learning and language use.
★ Machine learning	A branch of AI in which systems improve from data without being explicitly programmed.
Digital footprint	The trail of data a person leaves behind through their online activity.
Surveillance / tracking	Systematic monitoring of a person's activity, location or communications.

Part 2 — The Traps

This is where marks actually leak. Examiners deliberately test the pairs below, because most candidates can define one term and then blur the other.

Confusion Pairs You Will Be Tested On

Data vs Information

Data is raw facts with no context, e.g. 67, 82, 45. **Information** is data that has been processed and given meaning, e.g. "the class average is 64,7%".

Trap

Answering that "information is just more data". Your definition must contain the words **context, meaning or usefulness** to earn the mark.

Validation vs Verification

Validation asks whether the data is sensible, e.g. an age between 0 and 120. **Verification** asks whether the data matches the original source, e.g. typing a password twice.

Trap

Validation can never catch a correctly formatted wrong value. Entering age 25 instead of 52 passes validation and fails verification.

RAM (Random Access Memory) vs Virtual Memory vs Cache

Cache is smallest and fastest and sits on the CPU. RAM is the fast, volatile main working memory. Virtual memory is the slowest, lives on disk, and is only used once RAM is exhausted.

Trap

Saying that adding virtual memory speeds up a PC. It does the opposite — it is a slow fallback. The real solution is more RAM.

Virus vs Worm vs Trojan

A virus attaches to a host file and needs a user action to spread. A worm self-replicates across a network with no user help. A Trojan does nothing except pretend to be legitimate until it is opened.

Trap

Calling every threat "a virus". Use the specific term that the scenario points to.

Phishing vs Pharming

Phishing lures the user into clicking a fraudulent link. Pharming poisons the DNS or hosts file so that even the correct address takes the user to a fake site.

Trap

Pharming requires no user error at all, which is precisely why it is the more dangerous of the two.

Switch vs Router

A switch operates **within** a single network. A router operates **between** different networks.

Trap

A home "router" is actually a router, switch, access point and modem in one box. The exam wants each function described separately.

IP (Internet Protocol) Address vs MAC (Media Access Control) Address

An IP address is logical, usually assigned by DHCP (Dynamic Host Configuration Protocol), and changes when the device moves to another network. A MAC address is burnt into the NIC (Network Interface Card), permanent and globally unique.

Bandwidth vs Data Cap vs Throttling

Bandwidth is how **fast** (Megabits per second). A cap is how **much** (Gigabytes). Throttling is a deliberate reduction in speed, often applied once the cap is exceeded.

Trap

Being capped does not slow you down until you exceed the limit. Being shaped slows specific traffic types at any time.

PK (Primary Key) vs FK (Foreign Key)

A primary key uniquely identifies a record in its **own** table. A foreign key points at the primary key of **another** table in order to create the relationship.

Trap

One field can be both. Also remember: a primary key may never be null, but a foreign key may be.

Data Integrity vs Data Redundancy

Integrity means the data is correct. Redundancy means the data is repeated. Redundancy is what **causes** integrity failures, because one copy gets updated and the other does not.

Trap

Not all redundancy is bad — backups are deliberate redundancy. Read the context before condemning it.

Function vs Procedure

A function returns a value and can therefore be used inside an expression. A procedure performs an action and returns nothing.

Trap

If the question says "it displays something on screen", that is a procedure. Displaying is not returning.

Value vs Reference Parameters

A value parameter passes a **copy**, so changes disappear when the routine ends. A reference parameter passes the **actual variable**, so changes persist in the calling code.

Trap

Whenever a question says "the routine must change two values", you need reference parameters (or a function plus one reference parameter).

Syntax vs Logic vs Runtime Errors

A syntax error will not compile. A logic error compiles and runs but produces the wrong answer. A runtime error crashes the program during execution, e.g. division by zero or an array index out of bounds.

Trap

A logic error is the only kind a compiler will never find for you. That is exactly why trace tables exist.

Compiler vs Interpreter

A compiler translates the whole program in advance and produces a fast standalone executable. An interpreter translates and executes line by line, which is slower but easier to debug.

Alpha vs Beta Testing

Alpha testing is internal, carried out by the developers or the company. Beta testing is external, carried out by real users under real conditions.

Open Source vs Freeware

Open source means the **source code** is available and may be modified, and it may still cost money. Freeware means **free of charge**, but the source code is usually closed.

Trap

These are not synonyms. Free does not mean open.

Copyright vs DRM (Digital Rights Management)

Copyright is the **legal right** that protects the creator. DRM is the **technology** that enforces it — licence keys, activation limits, region codes, restrictions on printing or copying, and streaming services that block downloads.

Trap

DRM protects the distributor, not the buyer. It often restricts legal use too: you may be unable to make a lawful backup, move a purchased e-book to a new device, or play a legally bought disc in another region. If a question asks for a disadvantage of DRM, that is the answer it wants.

The Concepts Most Often Answered Badly

Normalisation

Do not simply say "removing duplicates". Say: restructuring tables so that every field depends on the whole primary key and on nothing else, which removes redundancy and prevents anomalies. If you are asked to justify it, name a specific anomaly:

- **Update anomaly:** a teacher's surname is stored in 200 rows; change one and the data becomes inconsistent.
- **Insert anomaly:** a new subject cannot be added until at least one learner enrolls for it.
- **Delete anomaly:** removing the last learner in a subject also erases the subject itself.

Data Warehouse vs Data Mining

The warehouse is the place: a large, read-only, historical, consolidated store. Mining is the process: finding hidden patterns inside it. The classic example is a supermarket mining loyalty-card data, discovering that nappies and energy drinks are bought together on Friday evenings, and repositioning the shelves accordingly.

Cloud Computing

Earn the marks by naming both sides. Advantages: accessible from anywhere, scalable on demand, no large capital outlay on hardware, automatic backups, easy collaboration. Disadvantages: useless without connectivity, ongoing subscription cost, privacy and data-sovereignty risk, vendor lock-in, and data caps.

Encapsulation

It is not merely "putting things into a class". It is hiding attributes as private and exposing them only through accessor and mutator methods, so that the object controls its own state and invalid data can be rejected before it is ever stored.

The Internet of Things

Always give a concrete example or you will lose the second mark: a smart geyser that reports its temperature to an app and switches itself off during peak tariff hours.

Green Computing

Examiners want more than "recycling". Cover the whole life cycle: energy-efficient hardware, sleep and hibernate settings, virtualisation to reduce the number of physical servers, duplex printing, responsible e-waste disposal, and extending device lifespan through repair and reuse.

The Digital Divide

It is not just "some people do not have computers". It is the gap in meaningful access to ICT (Information and Communication Technology), driven by the cost of devices and data, infrastructure and electricity availability, digital literacy, language, and disability.

POPIA (Protection of Personal Information Act)

Personal information must be collected lawfully for a stated purpose and with consent, kept only for as long as it is needed, kept accurate and secure, and the data subject may request access to it and correction of it.

Exam Technique — Non-Negotiables

- **Count the marks, then count your points.** A four-mark question needs four distinct facts, not one fact stated four ways.
- **"Motivate", "justify" and "explain why" mean the word *because* must appear.** A bare definition scores zero against those verbs.
- **"Give TWO examples" means give exactly two.** Markers assess the first two only, so a third guess cannot rescue you.
- **Never answer with a brand name where a term is wanted.** "Chrome" is not a definition of a browser.
- **Scenario questions demand scenario language.** If the stem is about a school library, your answer must mention learners, books or loans. Generic answers are penalised.
- **Split "advantages and disadvantages" answers visibly.** Two clean columns beat a paragraph the marker has to excavate.
- **Watch the negatives.** "Which of these is NOT a function of the operating system" catches hundreds of candidates every year.

Part 3 — Question 4 Masterclass: Databases

Question 4 of Paper 2 is Data and Information Management. It is one of the most predictable questions on the paper, because it recycles the same handful of skills: spot the design faults in a badly built table, propose the corrected structure, choose sensible data types, and draw or complete an ERD (Entity-Relationship Diagram). Learn the patterns below and this question becomes free marks.

What Question 4 almost always asks

Identify design errors in a given table and suggest corrections.

Split a flat table into properly normalised tables, naming the primary keys (PK) and foreign keys (FK).

Choose and justify the correct data type and field properties.

Draw or complete an ERD (Entity-Relationship Diagram), showing cardinality and the foreign key.

Explain a database concept in context: integrity, redundancy, an anomaly, a DBMS (Database Management System) advantage, or POPIA (Protection of Personal Information Act).

3.1 The Master Checklist of Database Design Errors

Run every table you are given through this list in order. Each item below has been examined repeatedly.

#	Design error	Why it is wrong and how to fix it
1	No primary key	Records cannot be uniquely identified and duplicates cannot be prevented. Add an autonumber ID field such as LearnerID.
2	Unsuitable primary key	Surname, e-mail or telephone number can repeat or change. Use a field that is unique, never null and never changes.
3	Non-atomic field	A field holding more than one value, e.g. "Thabo Mokoena" or "12 Main Rd, Bellville, 7530". Split into separate fields so that each can be sorted and searched independently.
4	Multiple values in one field	A field such as Subjects containing "IT, Maths, Physics". Impossible to query reliably. Move to a separate table linked by a foreign key.
5	Repeating groups	Columns such as Test1, Test2, Test3. A fourth test would require the table structure to change. Move to a separate table with one row per test result.
6	Calculated or derived field stored	Total, Age or VAT stored as data. It becomes stale and inconsistent. Calculate it in a query or in code when needed.
7	Redundant data	The teacher's name and e-mail repeated in every learner row. Wastes space and causes update anomalies. Move to its own table and link by ID.
8	Incorrect data type	Numbers stored as text or text stored as numbers. Prevents sorting, calculation or correct storage. See 3.2 below.
9	Field size wrong	A 5-character field for a surname truncates data; a 255-character field for a grade wastes space.
10	Data stored in field names	Columns named 2024, 2025, 2026. Adding a year means redesigning the table. Store the year as data in a field.
11	Two entities in one table	Learner details and subject details mixed together. Each table must describe exactly one entity.
12	Inconsistent data entry	"Cape Town", "cape town" and "CPT" in the same field. Use a lookup list or validation rule to restrict input.
13	No validation or required setting	Essential fields left blank or nonsense values accepted. Set required fields and add validation rules.
14	Descriptive data stored with the foreign key	Storing both TeacherID and TeacherSurname in the child table. Store only the foreign key; retrieve the rest through the relationship.
15	Ambiguous or inconsistent naming	Fields named "Name1" and "Nm" or a mixture of naming conventions. Use clear, consistent, descriptive names.

3.2 Data Types — The Ones Examiners Test

Field	Wrong choice	Correct choice	Reason
TelNumber	Number	Text	A leading zero is lost, and spaces, brackets and +27 cannot be stored. Telephone numbers are never used in calculations.
IDNumber	Number	Text	Thirteen digits with a possible leading zero, and no arithmetic is ever performed on it.
PostalCode	Number	Text	Leading zeros must be preserved.
DateOfBirth	Text	Date/Time	Cannot sort chronologically or calculate age from text.
Price / Fees	Text	Currency	Cannot be totalled or averaged, and the currency symbol should be formatting, not data.
IsPaid	Text ("Yes"/"No")	Boolean / Yes-No	Only two possible states; text invites inconsistent entry such as "y", "YES", "true".
Quantity	Real / Double	Integer	Whole items only; a fractional quantity is meaningless.
Average	Integer	Real / Double	Decimal precision is required.
Comments	Text (255)	Memo / Long text	Free-form feedback exceeds the short-text limit.
Province	Free-entry Text	Lookup list	A fixed set of nine values; a lookup enforces consistency.
Photo	Text	OLE (Object Linking and Embedding) Object / file path	An image is binary data, not a string.

Phrase your answer like this

"The field TelNumber should be of type **Text** and not **Number**, because a number field discards the leading zero and no calculations are ever performed on a telephone number."

Naming the field, naming both types and giving the reason is what secures all the marks. A bare "should be text" usually earns one mark out of two.

3.3 Worked Example A — The Flat Learner Table

This is the classic past-paper stimulus: a single table doing the work of four. Study the table, then the fault list.

The table as given: tblLearners

LearnerName	Grade	Subjects	Age	ClassTeacher	TeacherEmail	FeesOwing
Thabo Mokoena	12A	IT, Maths, Physics	17	Mrs P Naidoo	pnaidoo@school.co.za	R 1 250,00
Aisha Patel	12A	IT, Accounting	18	Mrs P Naidoo	pnaidoo@school.co.za	R 0,00
Sipho Dlamini	12B	Maths, Physics	17	Mr K de Beer	kdebeer@school.co.za	R 480,00
Aisha Patel	11C	IT, Maths	16	Mr K de Beer	kdebeer@school.co.za	R 95,50

The faults, in the order a marker expects them

Field	Error	Correction
(whole table)	No primary key. Note that "Aisha Patel" appears twice, so the name cannot identify a record.	Add an autonumber field LearnerID and set it as the primary key.
LearnerName	Not atomic: first name and surname share one field, so learners cannot be sorted by surname.	Split into FirstName and Surname.
Grade	Not atomic: the grade and the class group are combined in "12A".	Split into Grade (12) and ClassGroup (A), or store a ClassID foreign key.
Subjects	Multiple values in one field, and a repeating group in disguise. A learner's subjects cannot be counted or searched.	Create tblSubject and a junction table tblLearnerSubject holding LearnerID and SubjectID.
Age	Derived and volatile data. It is wrong within a year of capture.	Store DateOfBirth as a Date/Time field and calculate age when required.
ClassTeacher and TeacherEmail	Redundant data repeated in every row, causing an update anomaly if a teacher's address changes.	Create tblTeacher and store only TeacherID as a foreign key in the learner table.
FeesOwing	Stored as text with a currency symbol and spaces, so it cannot be totalled.	Change to Currency (or Double) and apply the currency symbol as formatting.

The corrected design

Marks are awarded for naming the tables, listing the fields, and clearly marking each primary key (PK) and foreign key (FK).

```
tblTeacher      ( TeacherID (PK), Title, Initials, Surname, Email )

tblLearner      ( LearnerID (PK), FirstName, Surname, DateOfBirth,
                  Grade, ClassGroup, FeesOwing, TeacherID (FK) )

tblSubject      ( SubjectID (PK), SubjectName )

tblLearnerSubject ( LearnerID (FK), SubjectID (FK) )
                  -- composite primary key: LearnerID + SubjectID
```

Relationships: one teacher registers many learners (1:M). One learner takes many subjects and one subject is taken by many learners (M:M), which is resolved by tblLearnerSubject.

3.4 Worked Example B — The Repeating Group

The second favourite. It looks harmless, which is exactly why candidates miss it.

The table as given: tblResults

LearnerName	Test1	Test2	Test3	Total	Average
Thabo Mokoena	45	52	48	145	48,3
Aisha Patel	61	58		119	59,5
Sipho Dlamini	38	41	44	123	41

The faults

1. **Repeating group.** Test1, Test2 and Test3 are the same attribute repeated. Adding a fourth test forces the table structure to change, and a learner who wrote only two tests leaves an empty field.
2. **Total and Average are derived fields.** They can be calculated from the marks and will become inconsistent the moment a mark is edited.
3. **No primary key, and the learner is identified by name.** Use a foreign key LearnerID instead.
4. **No context for the marks.** There is no way to record the subject, the date, the maximum mark or the type of assessment.
5. **Marks stored without a maximum.** A mark of 45 is meaningless unless the total for that test is known.

The corrected design

```
tblLearner ( LearnerID (PK), FirstName, Surname, Grade )
tblTest    ( TestID (PK), TestName, TestDate, MaxMark, SubjectID (FK) )
tblResult  ( ResultID (PK), LearnerID (FK), TestID (FK), MarkObtained )
```

Now any number of tests can be added without touching the structure, a learner who missed a test simply has no row, and the total and average are produced by a query using the SQL (Structured Query Language) aggregate functions SUM and AVG.

3.5 Worked Example C — Spot the Errors Yourself

Cover the answers and attempt this one first. It is written in the style of a six-mark "identify THREE errors and suggest a correction for each" question.

tblOrders

CustName	OrderDate	ItemsOrdered	Qty	PricePerItem	DeliveryAddress
J Smith	03/04/2026	Tonkotsu, Gyoza	2	R 95	14 Long St Cape Town 8001
j smith	2026-04-05	Miso Ramen	1,5	R 89	14 long street, cpt
P Nkosi	5 April 2026	Gyoza	3	R 45	8 Oak Ave Durban 4001

Answers

Field	Error	Correction
(whole table)	No primary key.	Add an autonumber OrderID as the primary key.
CustName	Customer details repeated and entered inconsistently ("J Smith" / "j smith"); not atomic.	Create tblCustomer with CustomerID, FirstName and Surname, and store CustomerID as a foreign key.
OrderDate	Stored as text, shown in three different formats, so it cannot be sorted chronologically.	Set the data type to Date/Time and apply a single input mask.
ItemsOrdered	More than one value in a single field, so items cannot be counted or totalled.	Create tblOrderItem with one row per item ordered, linked to tblMenuitem.
Qty	A fractional value (1,5) has been accepted for a whole item.	Set the data type to Integer and add a validation rule requiring a value greater than zero.
PricePerItem	Stored as text with a currency symbol; also redundant, since the price belongs to the menu item.	Store the price in tblMenuitem as Currency. Keep a copy in the order line only if historical pricing must be preserved.
DeliveryAddress	Not atomic and inconsistently captured; cannot be searched by suburb or postal code.	Split into StreetAddress, Suburb, City and PostalCode, and use a lookup for City.

Answering technique for "identify the errors"

The mark allocation tells you how many faults to find. Six marks usually means three errors plus three corrections, so always give both halves.

Name the field. "There is a mistake in the third column" earns nothing.

Use the formula: Field → what is wrong → what it should be → why. The "why" is frequently a separate mark.

Never list the same fault twice under two different field names unless the question asks for every occurrence.

If the question says "suggest a suitable primary key", justify it: unique, never null, never changing.

3.6 Drawing an ERD (Entity-Relationship Diagram) — The Reliable Method

An ERD question is worth marks for the entities, the primary and foreign keys, the relationship lines and the cardinality. Work through these steps in order and you will not lose them.

1. **Underline every noun in the scenario.** These are your candidate entities.
2. **Test each candidate.** If the school stores many instances of it, and each instance has several attributes of its own, it is an entity. If it holds a single value, it is an attribute. "Suburb" is normally an attribute; "Supplier" is normally an entity.
3. **Name each entity consistently** and in the singular, e.g. tblLearner, tblBook, tblLoan.
4. **Give every entity a primary key** and underline it in the diagram.
5. **Take each pair of entities and read the relationship in both directions.** "One learner writes many tests" and "one test is written by one learner" gives you 1:M.
6. **Place the foreign key on the "many" side.** This single rule is where most candidates lose marks.
7. **Resolve every many-to-many relationship with a junction table** containing the two foreign keys, usually as a composite primary key. A relational database cannot implement M:M directly.
8. **Label each relationship with a verb** such as "borrows", "belongs to" or "places".
9. **Check that every foreign key has the same data type as the primary key it refers to.**

Notation

CAPS accepts either crow's foot notation or the 1 and ∞ labels used in Microsoft Access. Choose one and be consistent throughout the diagram.

Cardinality	Access notation	Crow's foot notation
One-to-one	1 — 1	single bar at each end
One-to-many	1 — ∞	single bar at the "one" end, crow's foot at the "many" end
Many-to-many	∞ — ∞ (must be resolved)	crow's foot at both ends (must be resolved)
Optional participation	not shown	small circle next to the symbol (0..*)
Mandatory participation	not shown	bar next to the symbol (1..*)

Common ERD mistakes that cost marks

- **Putting the foreign key on the "one" side.** Ask yourself which record can only ever point to one other record — that record holds the foreign key.
- **Leaving a many-to-many relationship unresolved.** If you have drawn ∞ at both ends, you have not finished.
- **Copying the primary key into both tables.** The key travels in one direction only.
- **Drawing a relationship line where no linking field exists.** Every line must be justified by a foreign key.
- **Creating an entity for what is really an attribute,** e.g. a whole table for "Gender".
- **Forgetting the composite key on the junction table.**
- **Inventing entities the scenario never mentions.** Answer the scenario in front of you, not the one you revised.

Worked ERD 1 — The School Library

Scenario: A school library lends books to learners. A learner may borrow many books over the course of the year, and any book may be borrowed by many different learners over time. Every book belongs to exactly one category, and a category contains many books.

```
tblCategory ( CategoryID (PK), CategoryName )
    |
    | 1 : M   "contains"
    v
tblBook      ( BookID (PK), Title, Author, ISBN, CategoryID (FK) )
    |
    | 1 : M   "is borrowed in"
    v
tblLoan      ( LoanID (PK), LearnerID (FK), BookID (FK),
              DateBorrowed, DateDue, DateReturned )
    ^
    | 1 : M   "borrows"
    |
tblLearner   ( LearnerID (PK), FirstName, Surname, Grade )
```

Note the key insight: learners and books are in a many-to-many relationship, and tblLoan is the junction table that resolves it. It also carries its own attributes — the borrow, due and return dates — which is why it needs its own primary key rather than a composite of the two foreign keys.

Worked ERD 2 — The Restaurant Ordering System

Scenario: A restaurant takes orders from customers. A customer may place many orders. Each order consists of one or more items from the menu, and any menu item may appear on many different orders.

```
tblCustomer  ( CustomerID (PK), FirstName, Surname, Phone )
    |
    | 1 : M   "places"
    v
tblOrder     ( OrderID (PK), CustomerID (FK), OrderDate, TableNo )
    |
    | 1 : M   "consists of"
    v
tblOrderItem ( OrderID (FK), MenuItemID (FK), Quantity, PriceCharged )
              -- composite primary key: OrderID + MenuItemID
    ^
    | 1 : M   "appears on"
    |
tblMenuItem  ( MenuItemID (PK), ItemName, Description, CurrentPrice )
```

Why PriceCharged is stored in the line item: normally a derived or duplicated value would be a design fault, but a historical price is the exception. If the menu price later changes, old invoices must still reflect what the customer actually paid. If you use this in an answer, say so explicitly — that reasoning is worth a mark.

Part 4 — UML Class Diagrams and Object Classes

Object-class questions now appear in Section E of Paper 2 every year. The good news is that the scope is narrow: you are dealing with a single class in almost every case. Inheritance, composition and aggregation are not required. What is required is that you can read a class diagram fluently, recognise what kind of method you are looking at, and build a diagram from scratch when you are given nothing but a screenshot of an interface.

The three question shapes, and where they came from

Shape 1 — You are given a class diagram and asked questions about scope, method types and instantiation. See November 2024 Question 5.5 and May/June 2025 Question 5.1.

Shape 2 — You are given the Delphi class declaration instead of a diagram and asked the same kinds of questions. See November 2024 Question 5.5.

Shape 3 — You are given a screenshot of an application interface and asked to design the class diagram yourself. See November 2025 Question 5.3.3, worth 7 marks.

Every one of these is a single class. Not one of them asks for inheritance, composition, aggregation or association.

4.1 The Anatomy of a Class Diagram

A class diagram is a rectangle divided into three compartments, always in this order.

Compartment	Contents
1. Class name	The class name only, centred and in bold, and starting with T by Delphi convention, e.g. TSighting.
2. Attributes	One per line: visibility, then the field name, then a colon, then the data type.
3. Methods	One per line: visibility, then the method name, then any parameters in brackets, then a colon and the return type if it returns something.

Visibility symbols

Symbol	Meaning	In Delphi
– (minus)	Private	Declared under private. Accessible only from inside the class itself.
+ (plus)	Public	Declared under public. Accessible from any other form or class.

The rule that earns easy marks

Attributes are private. Methods are public. That is encapsulation in practice, and it is the pattern in every diagram the DBE has set. Show a public attribute and you will lose a mark.

Write it the way the DBE writes it

Textbooks often use pure UML notation with empty brackets after every method and commas between parameters. The DBE writes its class diagrams in a Delphi-flavoured style instead. Match the examiner:

Textbook UML style	DBE examination style
+ getName() : String	+ getName: String
+ setPosition(sPosition : String) : void	+ setPosition (sPosition: String)
+ Create(iID : Integer, sName : String)	+ constructor create (iID: Integer; sName: String)
– fName : String	- fName: String

- Parameters are separated by semicolons, exactly as in Delphi, not by commas.
- The word constructor is written out in full in front of the constructor name.
- A method that returns nothing simply has no colon and no return type after it.
- Attribute names carry the f prefix (for field) and parameter names carry a type prefix: i for Integer, s for String, b for Boolean, r for Real.

Either style will normally be accepted, but the DBE style costs you nothing and reads as more fluent. Be consistent within one answer.

4.2 The Four Kinds of Method — Learn This First

Almost every mark in this section depends on classifying a method correctly. There are only four categories, and the examiners test the boundaries between them relentlessly.

Category	How to recognise it	What it does and why
Constructor	Named create or Create, written after the keyword constructor.	Creates the object in memory and gives its attributes their starting values. It is the only way to instantiate an object.
Accessor (getter)	Name starts with get. Returns a value. Takes no parameter.	Returns the value of one private attribute so that code outside the class can read it.
Mutator (setter)	Name starts with set. Returns nothing. Always takes a parameter.	Changes the value of one private attribute. It needs the parameter because it must be told what the new value is.
Auxiliary (helper)	Everything else — toString, calculations, tests, formatting.	Performs processing on the object's data. This is where the actual work of the class happens.

Three examiner favourites, answered

Why does an accessor normally have no parameter? Because it only returns a value that the object already holds. No information has to be sent into the method for it to do its job.

Why must a mutator have a parameter? Because the method has to be told what the new value is. Without a parameter it would have nothing to assign.

Which methods are auxiliary? Any method that is not the constructor, not a get and not a set. In a typical diagram that means toString plus whatever calculation or test method the scenario needs — for example a method that works out whether an employee is allowed into a particular location.

4.3 Question Shape 1 — You Are Given the Diagram

This is the May/June 2025 Question 5.1 pattern. Work through the diagram below and answer the questions before reading the solutions. The questions are written in the same style and mark allocation as the real paper.

TDeliveryDriver
- fDriverID: Integer - fName: String - fVehicleType: String - fDeliveries: Integer
+ constructor create (iDriverID: Integer; sName: String) + getName: String + getDeliveries: Integer + setVehicleType (sVehicleType: String) + canDeliver (sArea: String): Boolean + toString: String

An object is declared in the main form as follows:

```
objDriver : TDeliveryDriver;
```

The questions

1. Explain why the line `objDriver.fName := 'Lerato';` in the main form will cause an error. (1)
2. State TWO ways in which the value of a private attribute can be changed from a different form or class. (2)
3. Explain why an accessor method normally does not have a parameter. (1)
4. Identify TWO auxiliary methods in the class diagram. (2)
5. The driver's name and whether he or she can deliver to a specific area must be displayed. Motivate why the `toString` method will not be used for this purpose. (2)
6. In the statement `objDriver := TDeliveryDriver.create(101, 'Lerato');`, briefly explain what `TDeliveryDriver` refers to. (1)
7. Write Delphi code for a method called `determineInitial` that returns the first letter of the driver's name. (2)

The answers

#	Model answer
1	fName is a private attribute, so it cannot be accessed directly from outside the class. Only the class itself may read or write it.
2	Use the mutator (set) method provided for that attribute, or pass the value as a parameter to the constructor when the object is instantiated. (A third acceptable answer in some memoranda is to change the attribute's scope to public, but that defeats encapsulation and should be named as a poor choice.)
3	Because it only returns a value that the object already stores. Nothing has to be passed into it for it to do its work.
4	canDeliver and toString. Neither is a constructor, an accessor or a mutator.
5	toString returns a general description of all of the object's attributes, whereas here only two specific pieces of information are needed. More importantly, the answer depends on which area is being asked about, so a parameter must be received — and toString takes no parameter. A separate method combining getName and canDeliver(sArea) should be written instead.
6	It refers to the object class itself — the blueprint or template from which the object is created. The constructor create is being called on that class in order to instantiate the objDriver object.
7	See the code below.

```
function TDeliveryDriver.determineInitial: String;
begin
    Result := fName[1];
end;

// Copy(fName, 1, 1) is equally acceptable.
```

A note on question 1

The May/June 2025 paper phrased this as an error "during runtime". In Delphi the compiler will actually flag it before the program runs. Answer the question as it is asked — the mark is for identifying that fName is private and therefore out of reach from outside the class. Do not spend time arguing about when the error occurs.

4.4 Question Shape 2 — You Are Given the Delphi Code

This is the November 2024 Question 5.5 pattern. Exactly the same knowledge is tested, but you must read a class declaration instead of a diagram.

```
type
  TTutor = class(TObject)
  private
    fTutorNo    : Integer;
    fName       : String;
    fAvailable  : Boolean;
    fNumLessons : Integer;
  public
    constructor Create(iTutorNo: Integer; sName: String;
                      iNumLessons: Integer);
    procedure setAvailable(bAvailable: Boolean);
    function  getName: String;
    function  toString: String;
  end;
```

The questions and answers

Question	Answer
Which keyword indicates that the attributes are not accessible outside the class? (1)	private
Which method is called to instantiate the object? (1)	The constructor, Create.
One accessor has already been declared. Write the declaration of another accessor that could be added. (2)	function getNumLessons: Integer; — or function getTutorNo: Integer; or function getAvailable: Boolean;
Identify a mutator method in the code and explain the purpose of a mutator. (2)	setAvailable is the mutator. Its purpose is to change or assign a new value to a private attribute from outside the class, since the attribute itself cannot be reached directly.

Where candidates throw away the accessor mark

The question asks for a **declaration**, which means the single line that belongs in the class definition — not the implementation with begin and end.

The declaration must start with function, because an accessor returns a value. Writing getNumLessons: Integer; on its own is incomplete.

The return type must match the attribute's data type. An accessor for fAvailable returns Boolean, not String.

Do not offer getName — the question says another accessor, and that one already exists.

4.5 Question Shape 3 — Design the Diagram from an Interface

This is the November 2025 Question 5.3.3 pattern, and it is the one worth the most marks. You are shown a screenshot of a data-capture form and asked to design the class diagram for the object behind it. The instruction lists exactly what must appear, and each requirement is worth marks, so treat that list as your checklist.

The method

1. **Read the interface and list every item of data the user captures.** Each input control on the form — every radio button group, spin edit, date picker, time picker, combo box or edit box — corresponds to one attribute.
2. **Give each attribute an f prefix, a sensible name and a data type** taken from the control that captures it. A DateTimePicker set to date mode gives a Date attribute; a spin edit gives an Integer; a radio group of names gives a String.
3. **Mark every attribute private with a minus sign** and every method public with a plus sign. The instruction usually awards a mark for showing accessibility.
4. **Write the constructor exactly as the instruction demands.** If it says "without parameters", write it with no brackets and no parameters, however unnatural that feels.
5. **Add the required number of accessors**, each returning the data type of its own attribute and taking no parameter.
6. **Add the required mutator with its parameter**, naming the parameter with the correct type prefix.
7. **Count your items against the instruction before moving on.** Five attributes means five, not four and not six.

Worked solution — the wildlife sighting object

The November 2025 interface captured the type of animal, the number of animals sighted, the location of the sighting, the date and the time. The instruction required five attributes, a constructor without parameters, two accessors, one mutator that receives a parameter, accessibility on everything and data types where necessary.

TSighting
- fAnimalType: String - fNumberSighted: Integer - fLocation: String - fDate: Date - fTime: Time
+ constructor create + getAnimalType: String + getNumberSighted: Integer + setLocation (slocation: String) + toString: String

How the seven marks are earned

Requirement	Marks	Check
Five attributes, correctly named and matching the data captured on the interface	2	
All attributes marked private with a minus sign	1	
A constructor method shown with no parameters	1	
Two accessor methods, each with the correct return type	1	
One mutator method that receives a parameter of the correct type	1	
All methods marked public with a plus sign	1	
Total	7	

That breakdown is a reasonable reading of the mark allocation rather than the published memorandum, but the requirement list in the question is official — tick off each bullet and the marks follow.

Traps in the design question

Do not add extra attributes for things the interface does not capture. A caption, a heading or a button is not data.

Buttons are not methods of the object. "Record sighting" and "Clear" belong to the form, not to the TSighting class.

A constructor "without parameters" still has to initialise the attributes, usually to empty strings and zeros. The diagram shows only the header, but be ready to explain this.

Date and Time are separate attributes when the interface has separate pickers for each. Do not merge them into one.

toString counts as an auxiliary method, not an accessor, so it does not satisfy a "two accessor methods" requirement.

Give data types for every attribute. "Data types where necessary" means everywhere they are not obvious, which in practice means everywhere.

4.6 Writing the Class in Delphi

If you are asked to convert a diagram into a declaration, or to write a short method, this is the full pattern. Note how each line of the diagram maps onto exactly one line of code.

```
unit uSighting;

interface

uses SysUtils;

type
  TSighting = class(TObject)
  private
    fAnimalType    : String;
    fNumberSighted : Integer;
    fLocation      : String;
    fDate          : TDate;
    fTime          : TTime;
  public
    constructor Create;
    function getAnimalType: String;
    function getNumberSighted: Integer;
    procedure setLocation(sLocation: String);
    function toString: String;
  end;

implementation

constructor TSighting.Create;
begin
  fAnimalType    := '';
  fNumberSighted := 0;
  fLocation      := '';
  fDate          := Date;
  fTime          := Time;
end;

function TSighting.getAnimalType: String;
begin
  Result := fAnimalType;
end;

procedure TSighting.setLocation(sLocation: String);
begin
  fLocation := sLocation;
end;

function TSighting.toString: String;
begin
  Result := IntToStr(fNumberSighted) + ' x ' + fAnimalType +
    ' at ' + fLocation + ' on ' +
    DateToStr(fDate) + ' at ' + TimeToStr(fTime);
end;

end.
```

Short methods you should be able to write on demand

Two-mark method questions are common. They always follow the same shape: a header line and a single assignment to Result.

```
// Return the first letter of a name
function TSighting.determineInitial: String;
begin
    Result := fAnimalType[1];
end;

// Return whether a condition is met
function TSighting.isRareSighting: Boolean;
begin
    Result := fNumberSighted = 1;
end;

// Return a calculated value
function TSighting.calculateAverage(iDays: Integer): Double;
begin
    Result := fNumberSighted / iDays;
end;
```

The three ways two-mark method questions are lost

Leaving out function or writing procedure when a value must be returned.

Forgetting Result := and instead writing something like ShowMessage. Returning is not the same as displaying, and the question almost always says the value must be returned to the main form.

Omitting the class name in the implementation header. It is function TSighting.determineInitial, not function determineInitial.

4.7 The Model Answer Bank

These stems recur almost verbatim. Learn the wording of the answers, not just the idea.

If the question asks...	Write something like...
What does the minus sign indicate?	That the attribute is private and can only be accessed from within the class itself.
What does the plus sign indicate?	That the method is public and can be called from another form or class.
Which keyword hides the attributes?	private
Which method instantiates the object?	The constructor, Create.
What does the class name in TThing.create refer to?	The object class itself — the blueprint from which the object is created.
Why can a private attribute not be assigned directly?	Because private members are only visible inside the class; code in the main form has no access to them.
Name two ways to change a private attribute from outside.	Call the mutator method for that attribute, or pass the value as a parameter to the constructor.
Why does an accessor have no parameter?	Because it merely returns a value the object already holds; nothing needs to be sent in.
What is the purpose of a mutator?	To assign a new value to a private attribute from outside the class in a controlled way.
Identify the auxiliary methods.	Any method that is not the constructor, not a get and not a set — typically toString plus a calculation or test method.
What is the purpose of toString?	To return a single formatted string describing the current values of the object's attributes, for display in the main form.
Why is toString unsuitable here?	Because only specific information is required, and the information depends on a value that must be passed in as a parameter, which toString does not accept.
Why are attributes made private?	To apply encapsulation: the object controls its own data, and invalid values can be rejected by the mutator before being stored.

4.8 Diagram to Delphi Quick Reference

In the class diagram	In the Delphi class declaration
- fName: String	fName : String; (under private)
- fCount: Integer	fCount : Integer; (under private)
+ constructor create	constructor Create; (under public)
+ constructor create (iID: Integer)	constructor Create(iID: Integer);
+ getName: String	function getName: String;
+ getCount: Integer	function getCount: Integer;
+ setName (sName: String)	procedure setName(sName: String);
+ isValid: Boolean	function isValid: Boolean;
+ calcTotal (iQty: Integer): Real	function calcTotal(iQty: Integer): Double;
+ toString: String	function toString: String;
Final checklist before you leave this question Every attribute has a minus sign, a name and a data type. Every method has a plus sign. There is exactly one constructor. Every get returns something; no get takes a parameter. Every set takes a parameter; no set returns anything. You have written the number of items the question asked for — no more and no fewer.	

Part 5 — Abbreviation Rapid Recall

Cover the right-hand column and work down the list. Every one of these has been examined as a "what does the abbreviation stand for" question or has appeared inside a longer answer where the full form was expected.

One warning

Wi-Fi is a trade name registered by the Wi-Fi Alliance and is not an abbreviation of anything. If a question lists it among abbreviations, say so — do not invent "Wireless Fidelity".

Abbr.	Stands for	Abbr.	Stands for
2FA	Two-Factor Authentication	LTE	Long Term Evolution
5G	fifth generation (cellular)	MAC	Media Access Control
AI	Artificial Intelligence	MAN	Metropolitan Area Network
BIOS	Basic Input/Output System	Mbps	Megabits per second
BYOD	Bring Your Own Device	MFA	Multi-Factor Authentication
CMS	Content Management System	NFC	Near Field Communication
CPU	Central Processing Unit	NIC	Network Interface Card
CSS	Cascading Style Sheets	OLAP	Online Analytical Processing
CSV	Comma-Separated Values	OLE	Object Linking and Embedding
CVS	Computer Vision Syndrome	OS	Operating System
DBMS	Database Management System	P2P	Peer-to-Peer
DDoS	Distributed Denial of Service	PaaS	Platform as a Service
DHCP	Dynamic Host Configuration Protocol	PAN	Personal Area Network
DNS	Domain Name System	PIN	Personal Identification Number
DRM	Digital Rights Management	PK	Primary Key
ECT Act	Electronic Communications and Transactions Act	POP3	Post Office Protocol version 3
EFT	Electronic Funds Transfer	POPIA	Protection of Personal Information Act
ERD	Entity-Relationship Diagram	RAID	Redundant Array of Independent Disks
EULA	End User Licence Agreement	RAM	Random Access Memory
FK	Foreign Key	RFID	Radio Frequency Identification
FTP	File Transfer Protocol	ROM	Read-Only Memory
GB	Gigabyte	RSI	Repetitive Strain Injury
GHz	Gigahertz	RSS	Really Simple Syndication
GIGO	Garbage In, Garbage Out	SaaS	Software as a Service
GPS	Global Positioning System	SDLC	Software Development Life Cycle
GPU	Graphics Processing Unit	SMTP	Simple Mail Transfer Protocol
GUI	Graphical User Interface	SQL	Structured Query Language
HDD	Hard Disk Drive	SSD	Solid State Drive
HTML	HyperText Markup Language	TCP	Transmission Control Protocol
HTTP	Hypertext Transfer Protocol	TLS	Transport Layer Security
HTTPS	Hypertext Transfer Protocol Secure	UEFI	Unified Extensible Firmware Interface
IaaS	Infrastructure as a Service	UML	Unified Modeling Language
ICT	Information and Communication Technology	UPS	Uninterruptible Power Supply
IDE	Integrated Development Environment	URL	Uniform Resource Locator
IMAP	Internet Message Access Protocol	UTP	Unshielded Twisted Pair
IoT	Internet of Things	VoIP	Voice over Internet Protocol
IP	Internet Protocol	VPN	Virtual Private Network
ISP	Internet Service Provider	WAN	Wide Area Network

Abbr.	Stands for	Abbr.	Stands for
JSON	JavaScript Object Notation	WAP	Wireless Access Point
LAN	Local Area Network	XML	eXtensible Markup Language